IMU Dataset SimCLR Supervised Results

Below are the results for the experiments running a trained SimCLR encoder (via unsupervised learning), and gauging its performance across various downstream supervised tasks.

This results file is split by task (via “ClassNum”, then using linear/MLP probe, and shows the performance across various sizes)

Task 1: ClassNum 2

Binary Classification: Are you touching your face?

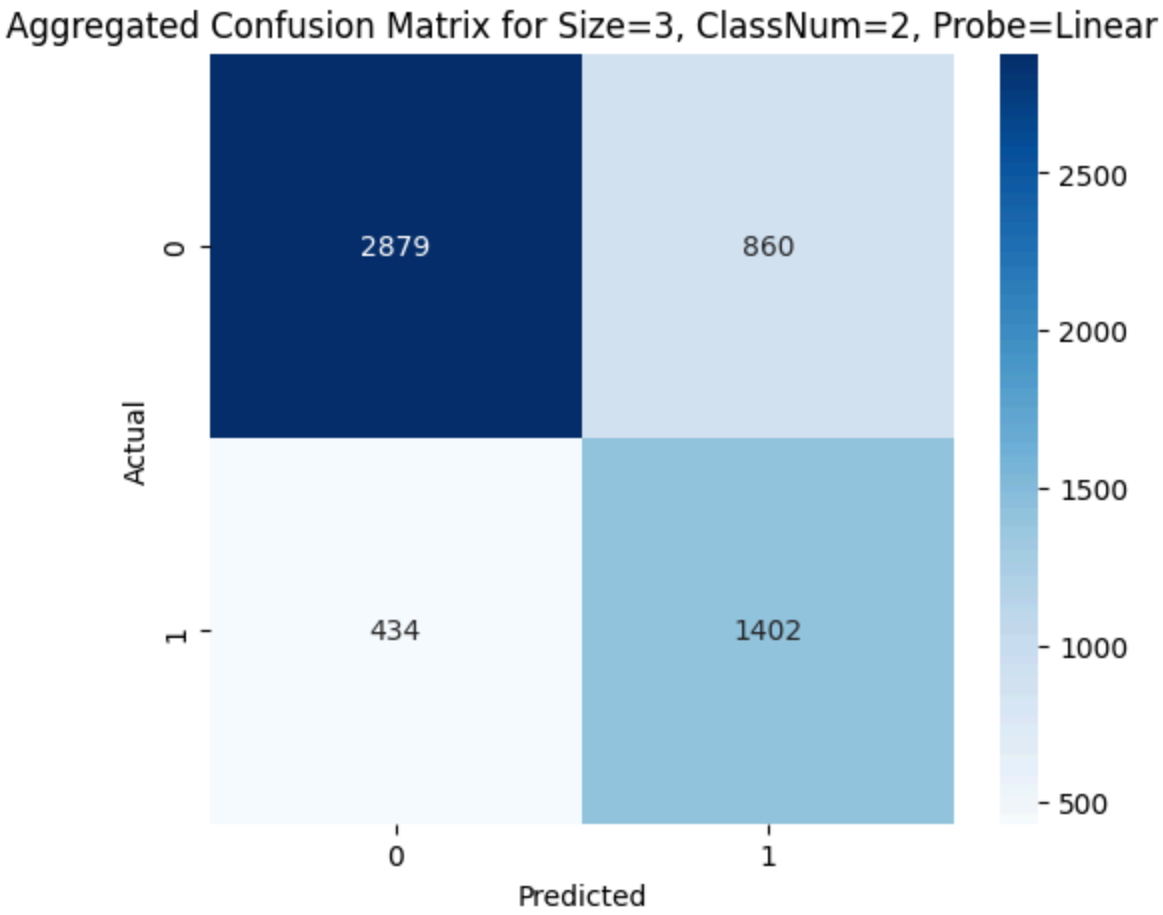
Linear Probe

Size: 3

Train	Validation	Test
Accuracy: 0.8320 Loss: 0.3786	Accuracy: 0.7726 Loss: 0.4886	Accuracy: 0.7668 Loss: 0.4769

Precision	Recall	F1 Score
0.6185	0.7620	0.6826

Confusion Matrix:

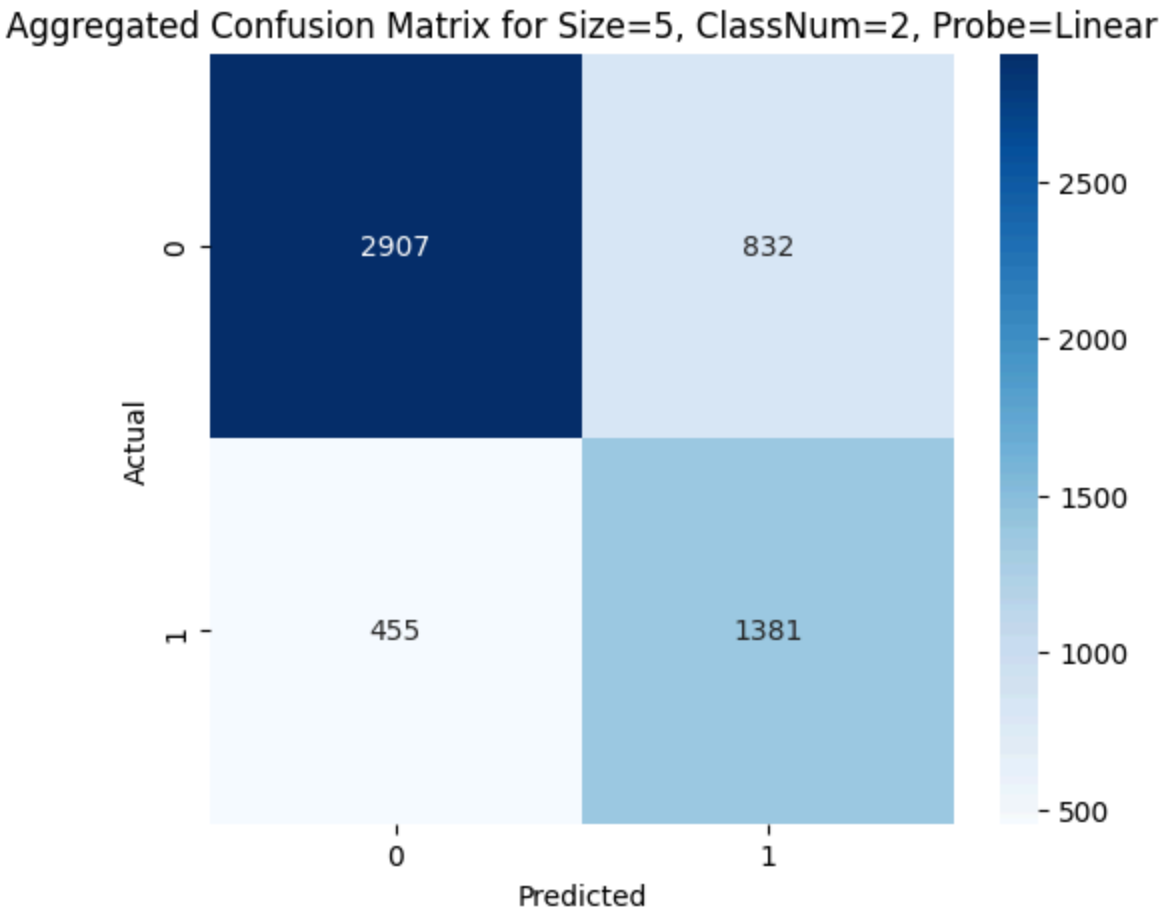


Size: 5

Train	Validation	Test
Accuracy: 0.8265 Loss: 0.3905	Accuracy: 0.7609 Loss: 0.4827	Accuracy: 0.7685 Loss: 0.4693

Precision	Recall	F1 Score
0.6232	0.7538	0.6819

Confusion Matrix:

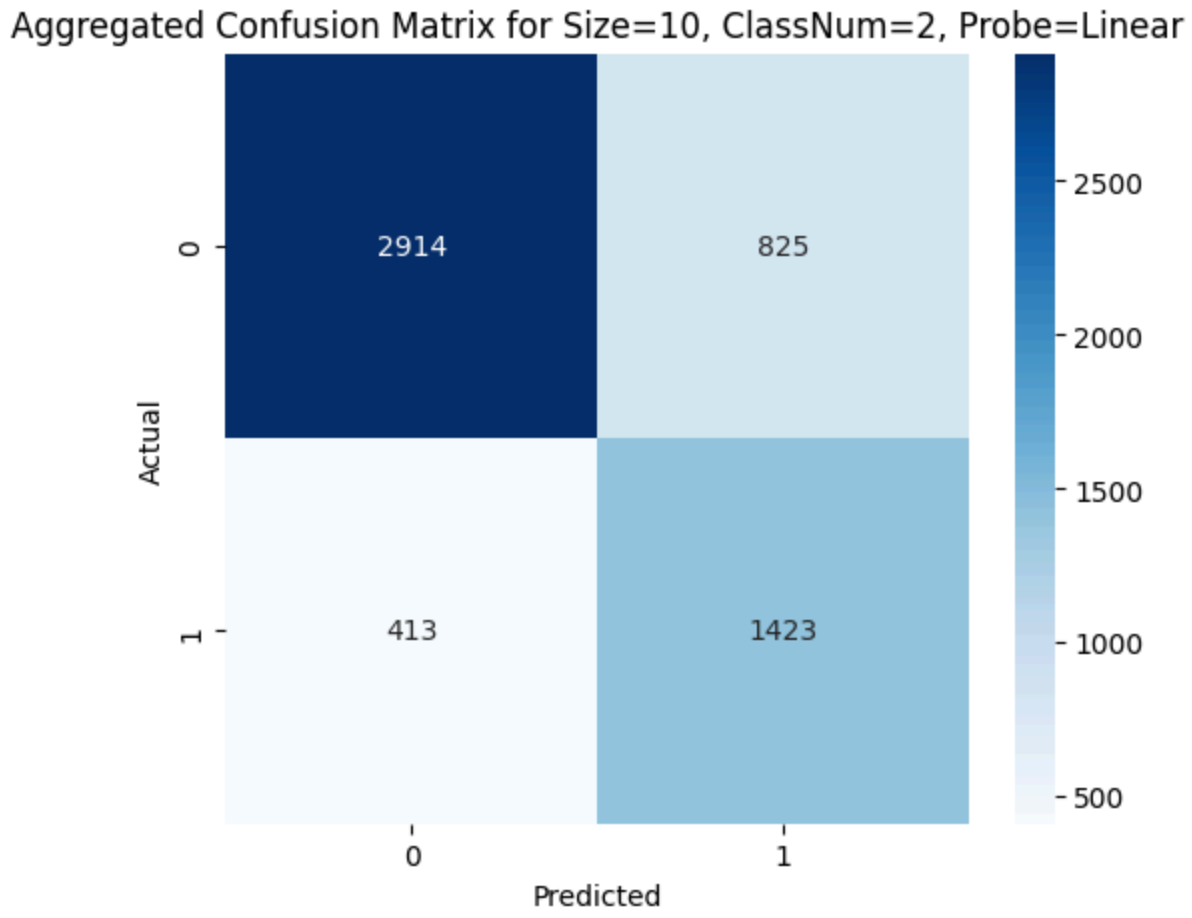


Size: 10

Train	Validation	Test
Accuracy: 0.8352 Loss: 0.3715	Accuracy: 0.7749 Loss: 0.4801	Accuracy: 0.7778 Loss: 0.4637

Precision	Recall	F1 Score
0.6333	0.7761	0.6970

Confusion Matrix:

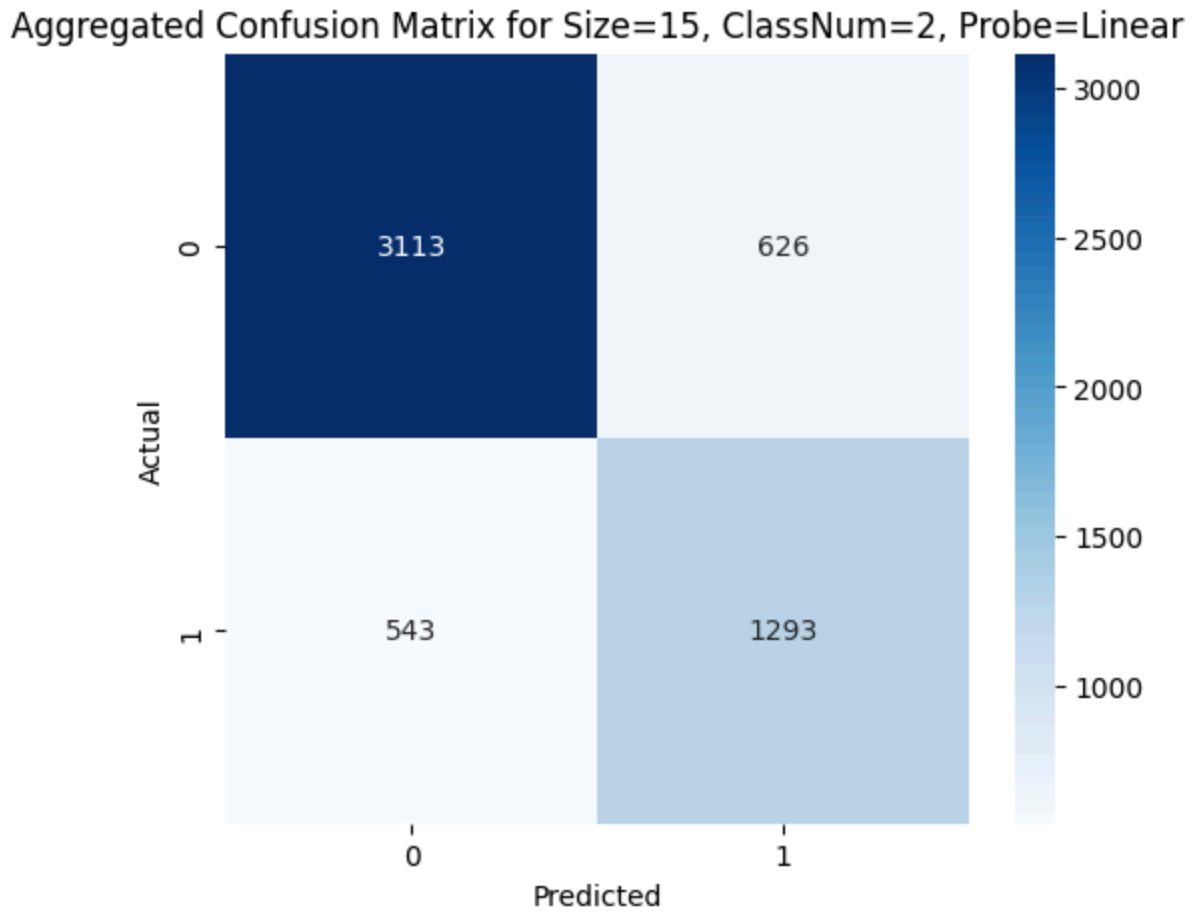


Size: 15

Train	Validation	Test
Accuracy: 0.8326 Loss: 0.3703	Accuracy: 0.7870 Loss: 0.4532	Accuracy: 0.7892 Loss: 0.4352

Precision	Recall	F1 Score
0.6720	0.7048	0.6877

Confusion Matrix:

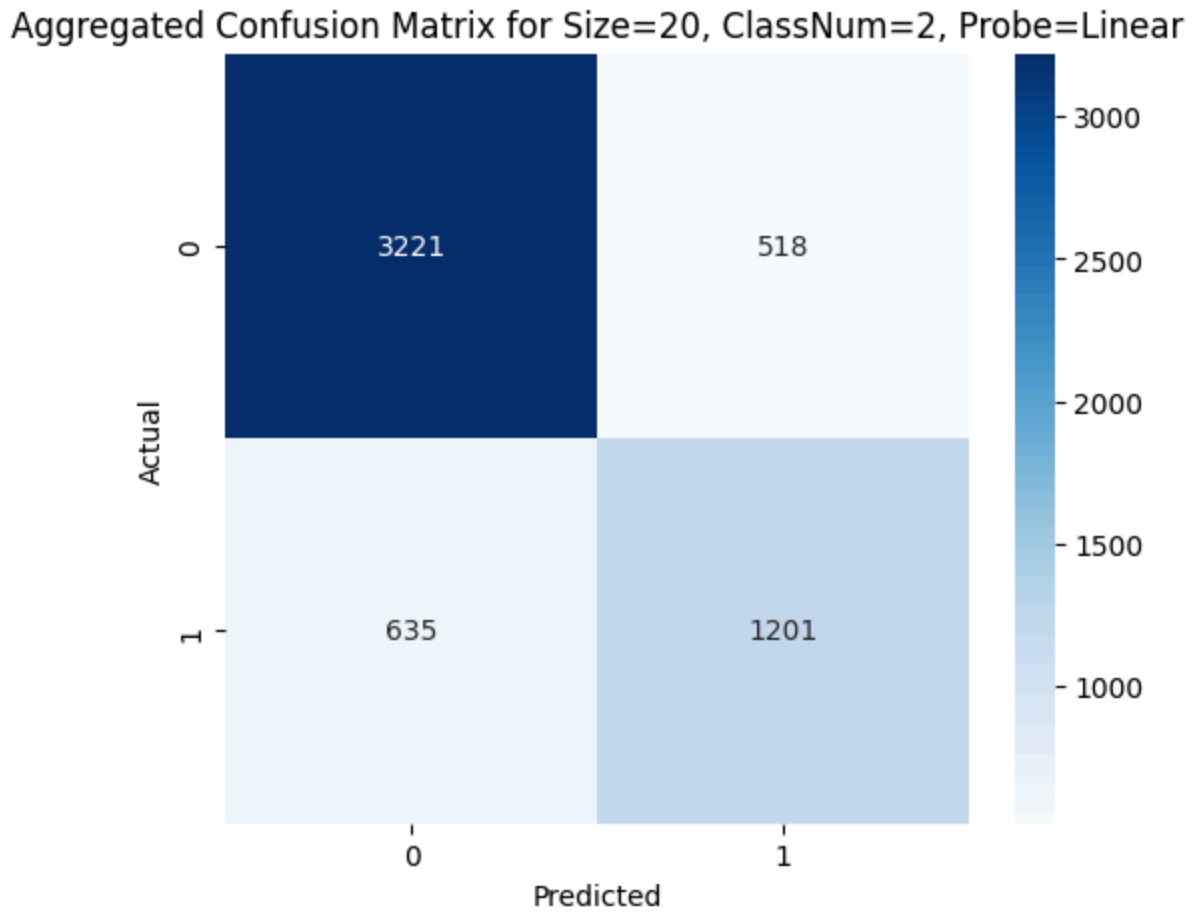


Size: 20

Train	Validation	Test
Accuracy: 0.8401 Loss: 0.3564	Accuracy: 0.7911 Loss: 0.4441	Accuracy: 0.7921 Loss: 0.4292

Precision	Recall	F1 Score
0.6971	0.6555	0.6751

Confusion Matrix:



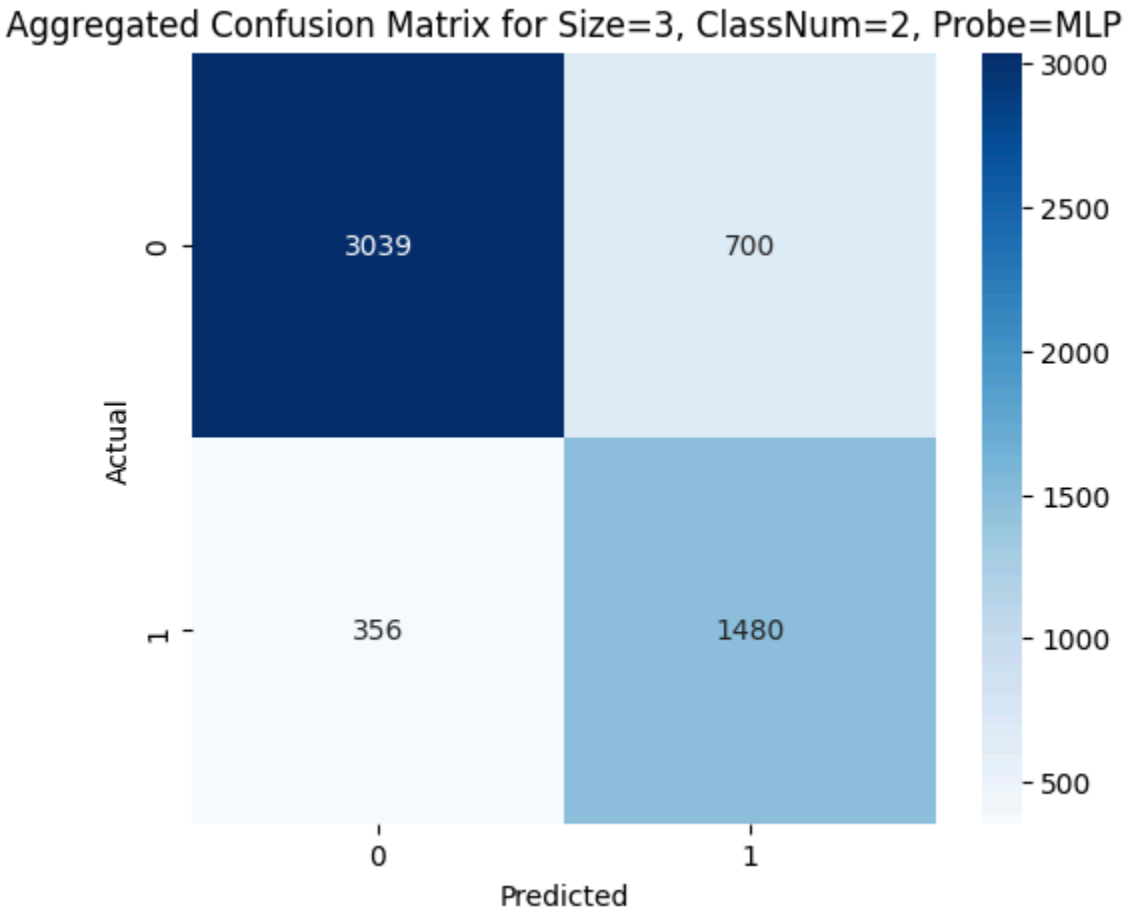
MLP Probe

Size: 3

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0031	Accuracy: 0.8017 Loss: 0.8019	Accuracy: 0.8091 Loss: 0.7487

Precision	Recall	F1 Score
0.6775	0.8083	0.7366

Confusion Matrix:

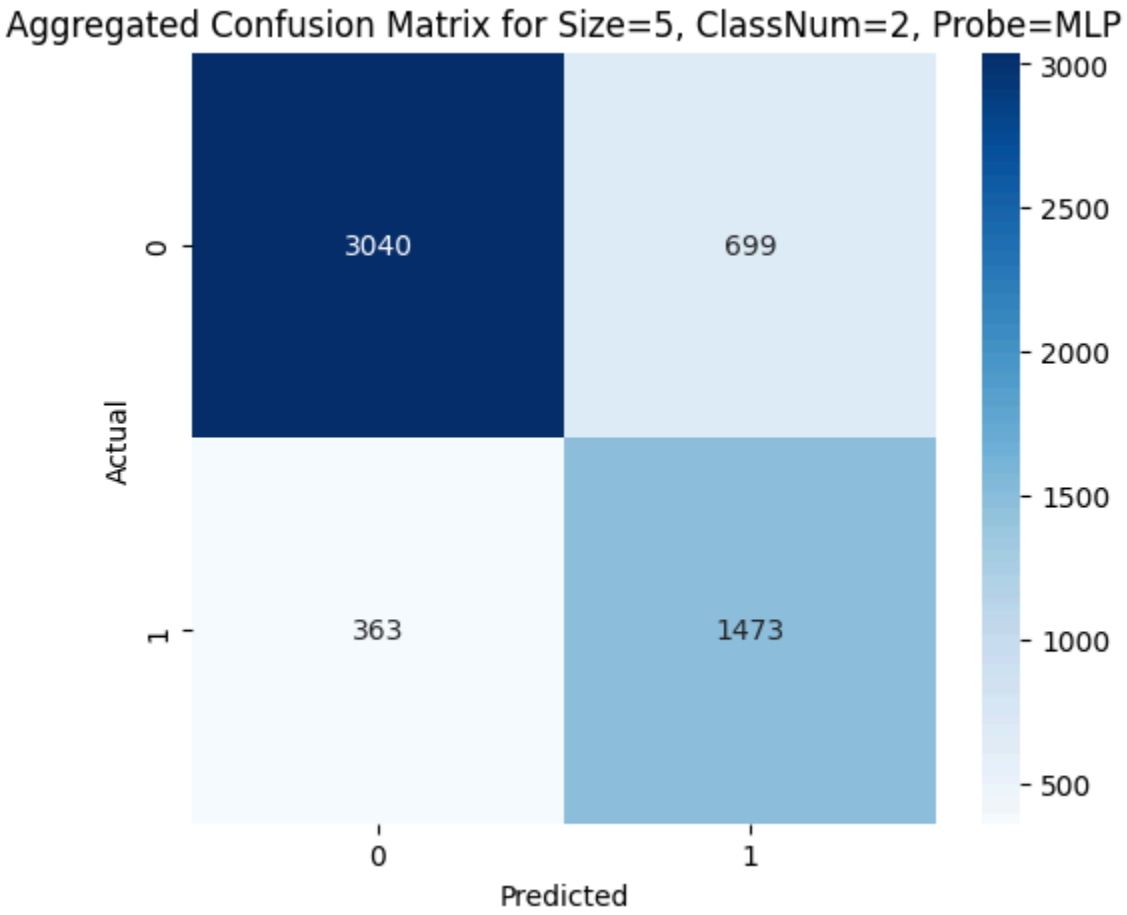


Size: 5

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0025	Accuracy: 0.8041 Loss: 0.8555	Accuracy: 0.8087 Loss: 0.7991

Precision	Recall	F1 Score
0.6779	0.8039	0.7351

Confusion Matrix:

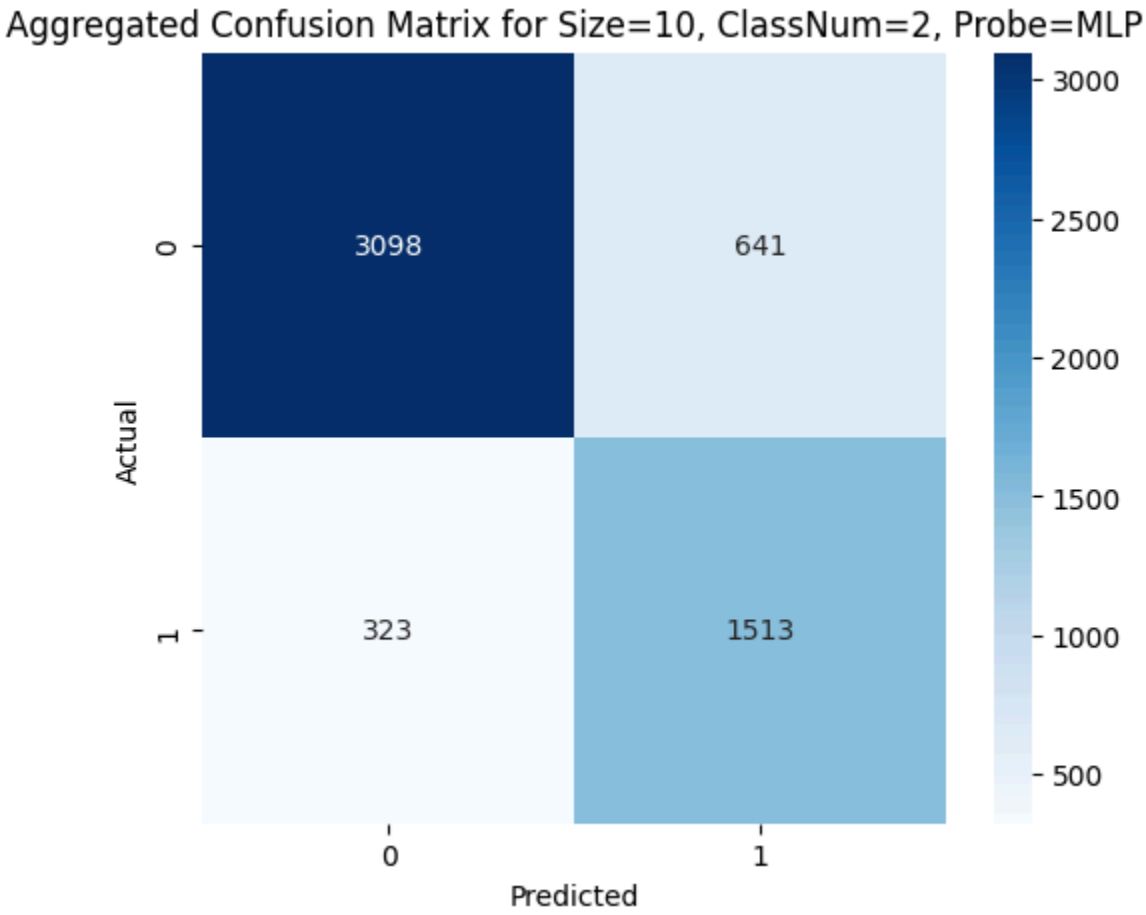


Size: 10

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0019	Accuracy: 0.8206 Loss: 0.9075	Accuracy: 0.8262 Loss: 0.8307

Precision	Recall	F1 Score
0.7021	0.8251	0.7580

Confusion Matrix:

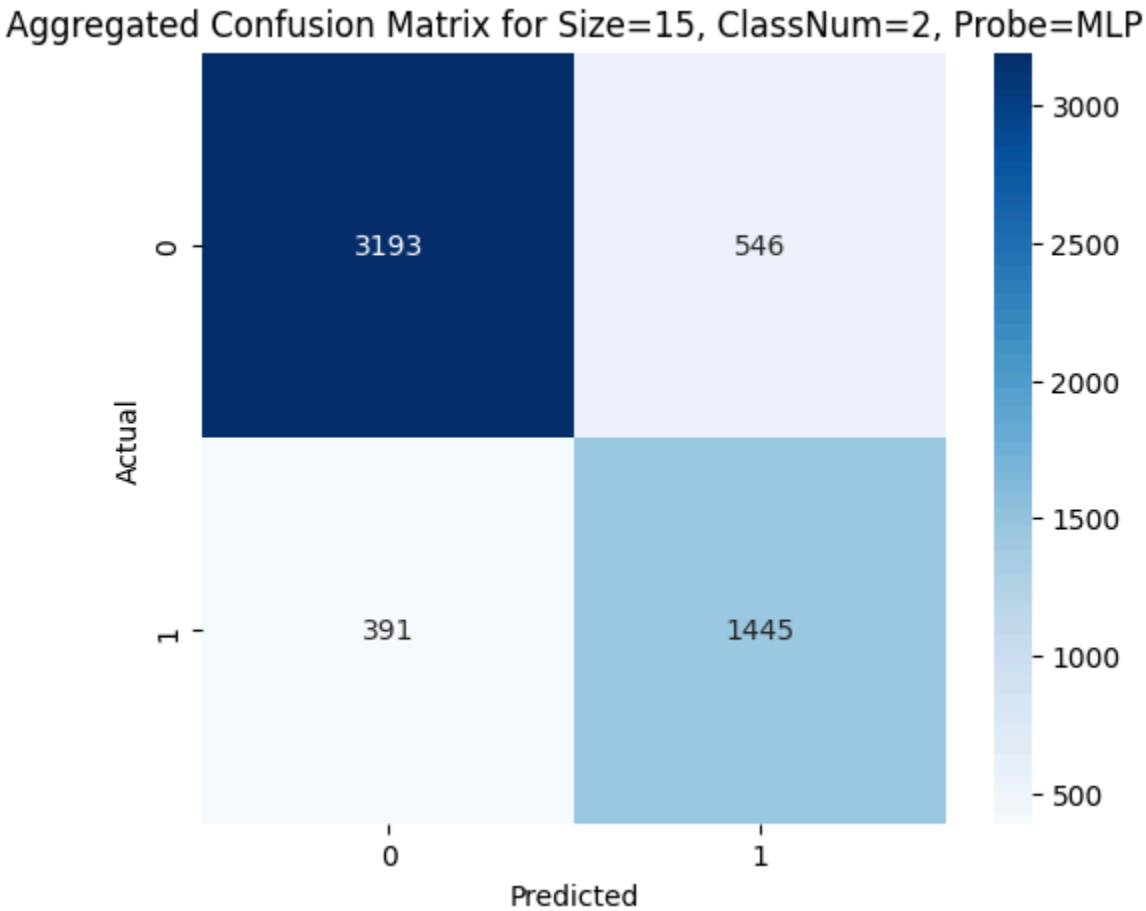


Size: 15

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0016	Accuracy: 0.8260 Loss: 0.9053	Accuracy: 0.8314 Loss: 0.8249

Precision	Recall	F1 Score
0.7261	0.7894	0.7557

Confusion Matrix:

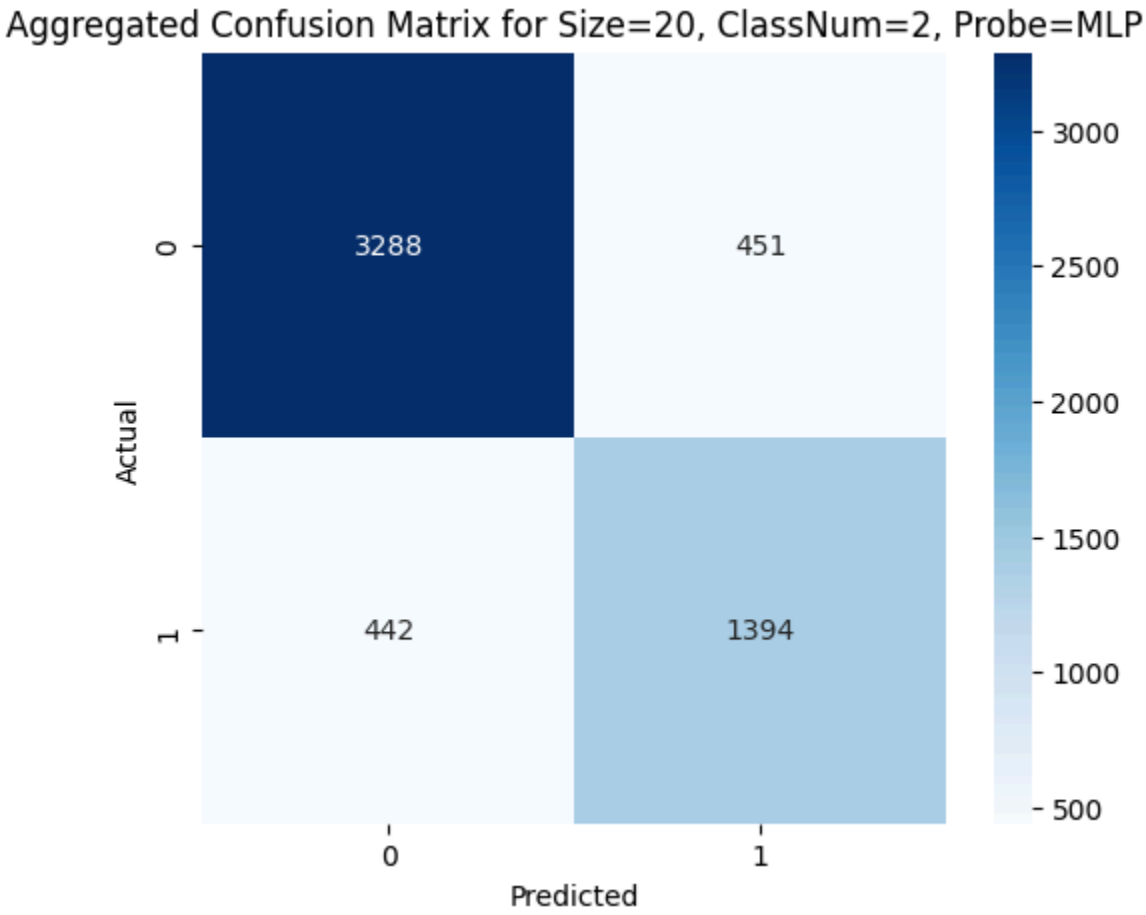


Size: 20

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0014	Accuracy: 0.8327 Loss: 0.8750	Accuracy: 0.8395 Loss: 0.8304

Precision	Recall	F1 Score
0.7563	0.7607	0.7579

Confusion Matrix:



Task 2: ClassNum 3

Touch type classification: Brief touch vs. Continuous touch vs. No touch

Linear Probe

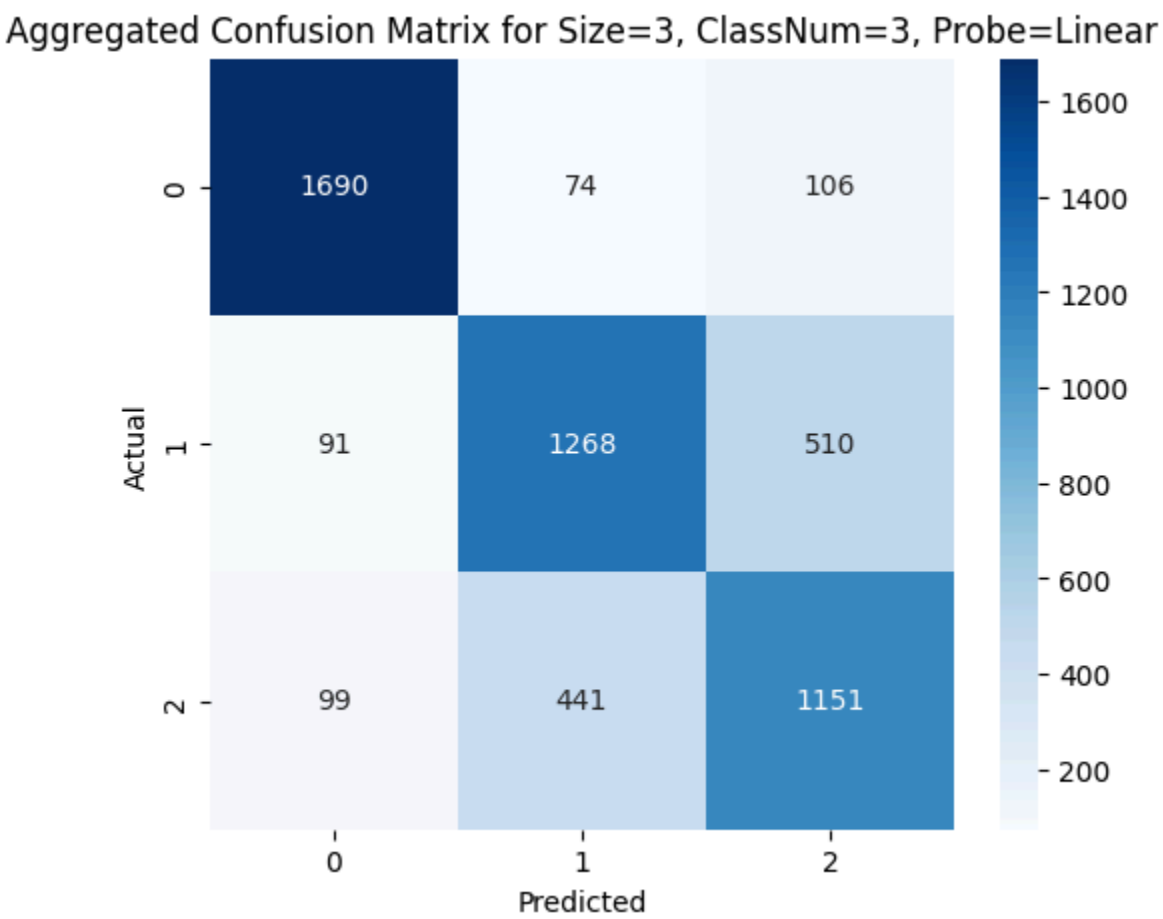
Size: 3

Train	Validation	Test

Accuracy: 0.8529 Loss: 0.3846	Accuracy: 0.7569 Loss: 0.5723	Accuracy: 0.7530 Loss: 0.5847

Precision	Recall	(Macro) F1 Score
0.7516	0.7508	0.7498

Confusion Matrix:



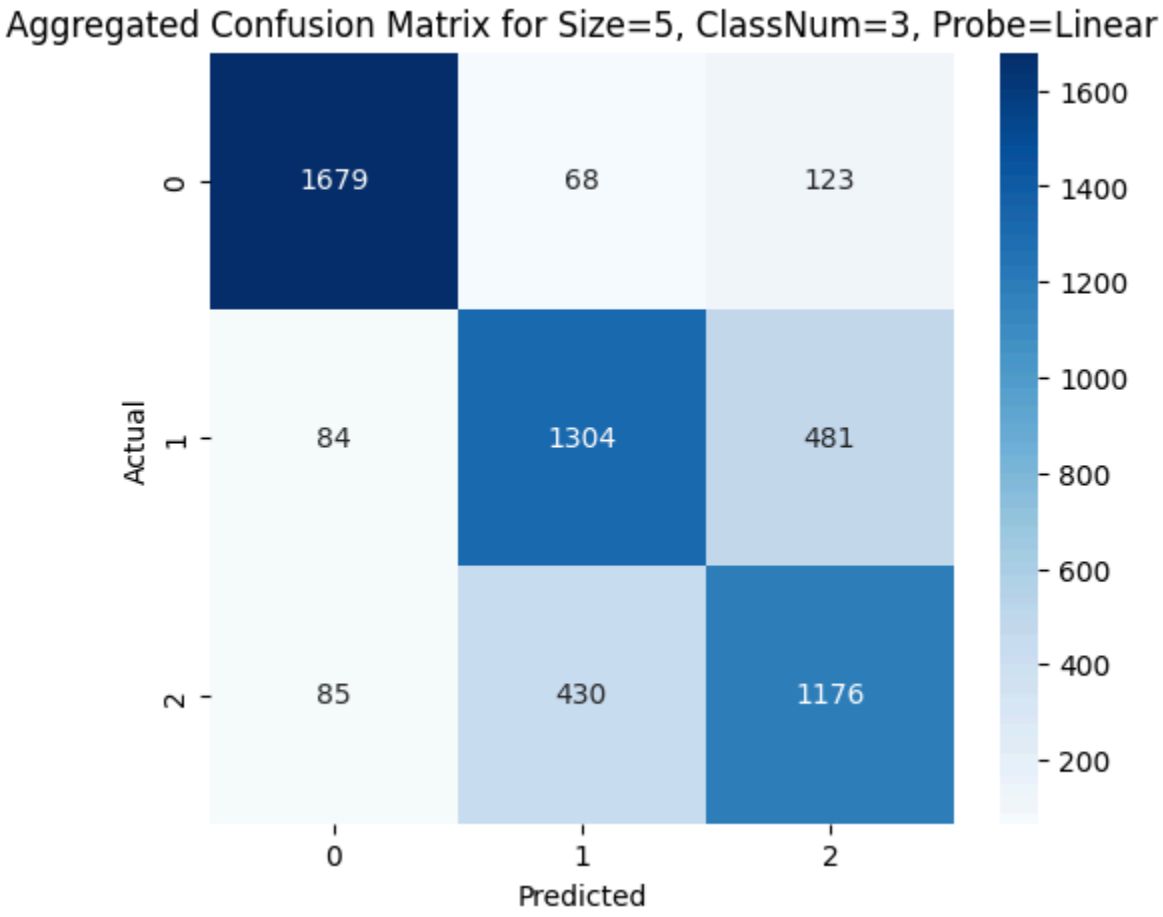
Size: 5

Train	Validation	Test

Accuracy: 0.8447 Loss: 0.3916	Accuracy: 0.7695 Loss: 0.5438	Accuracy: 0.7639 Loss: 0.5530

Precision	Recall	(Macro) F1 Score
0.7647	0.7620	0.7614

Confusion Matrix:



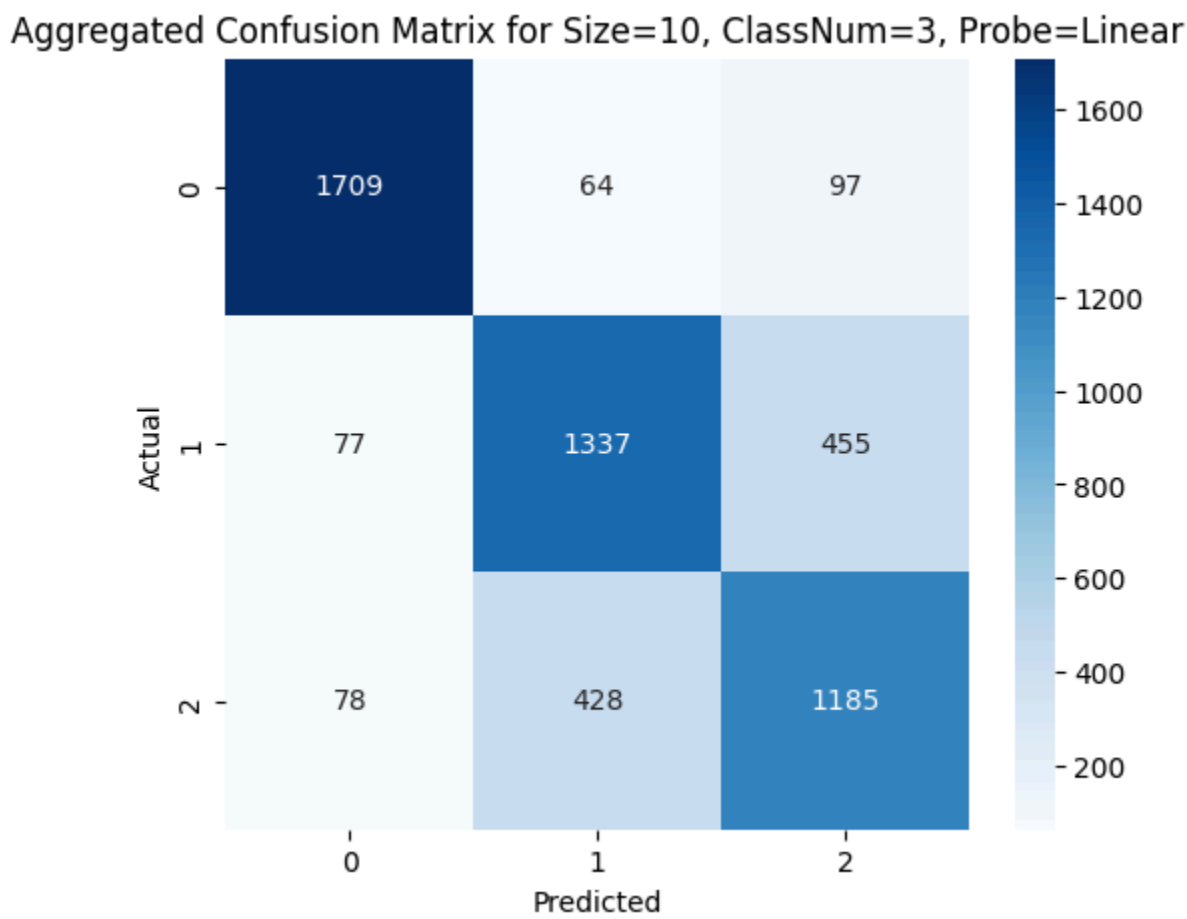
Size: 10

Train	Validation	Test

Accuracy: 0.8440 Loss: 0.3974	Accuracy: 0.7781 Loss: 0.5107	Accuracy: 0.7761 Loss: 0.5233

Precision	Recall	(Macro) F1 Score
0.7762	0.7739	0.7728

Confusion Matrix:



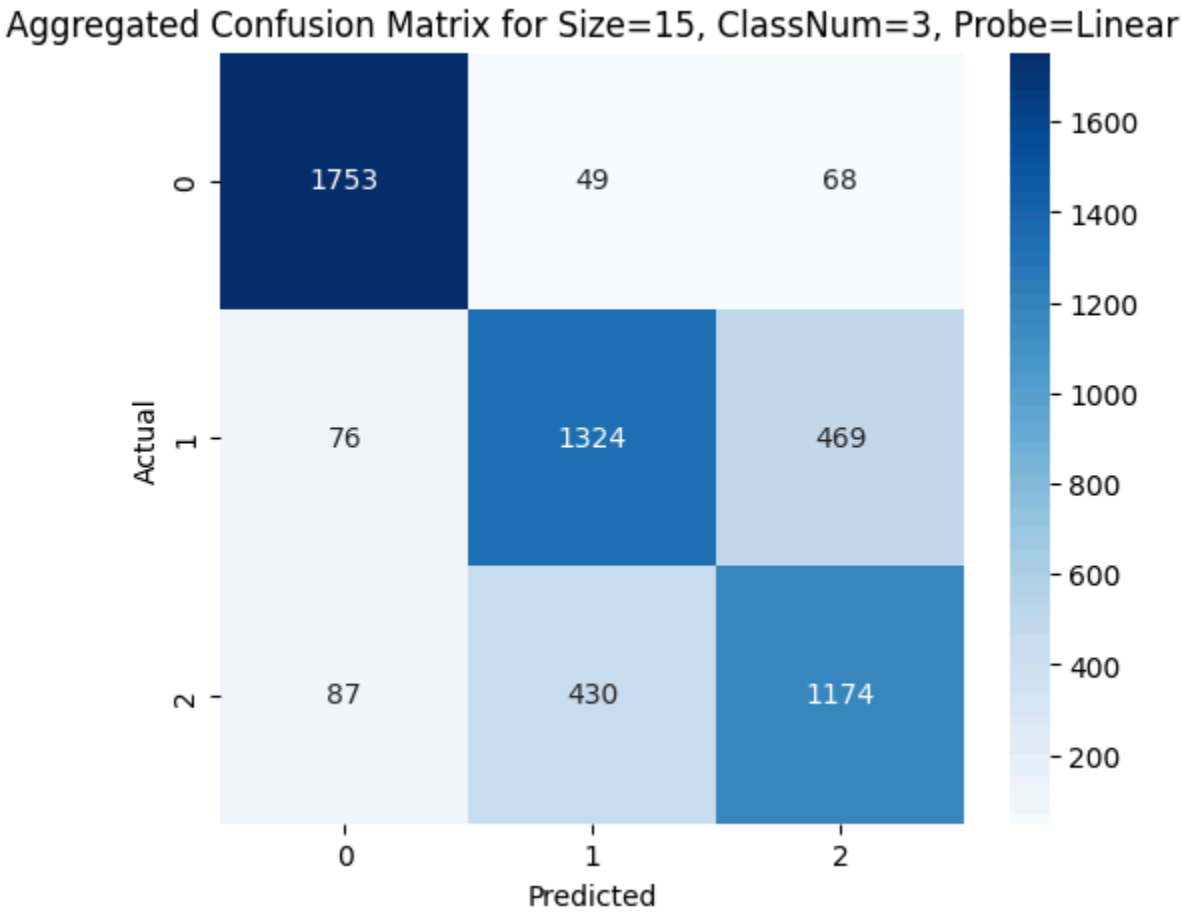
Size: 15

Train	Validation	Test

Accuracy: 0.8399 Loss: 0.3862	Accuracy: 0.7851 Loss: 0.4938	Accuracy: 0.7802 Loss: 0.5144

Precision	Recall	(Macro) F1 Score
0.7784	0.7777	0.7755

Confusion Matrix:

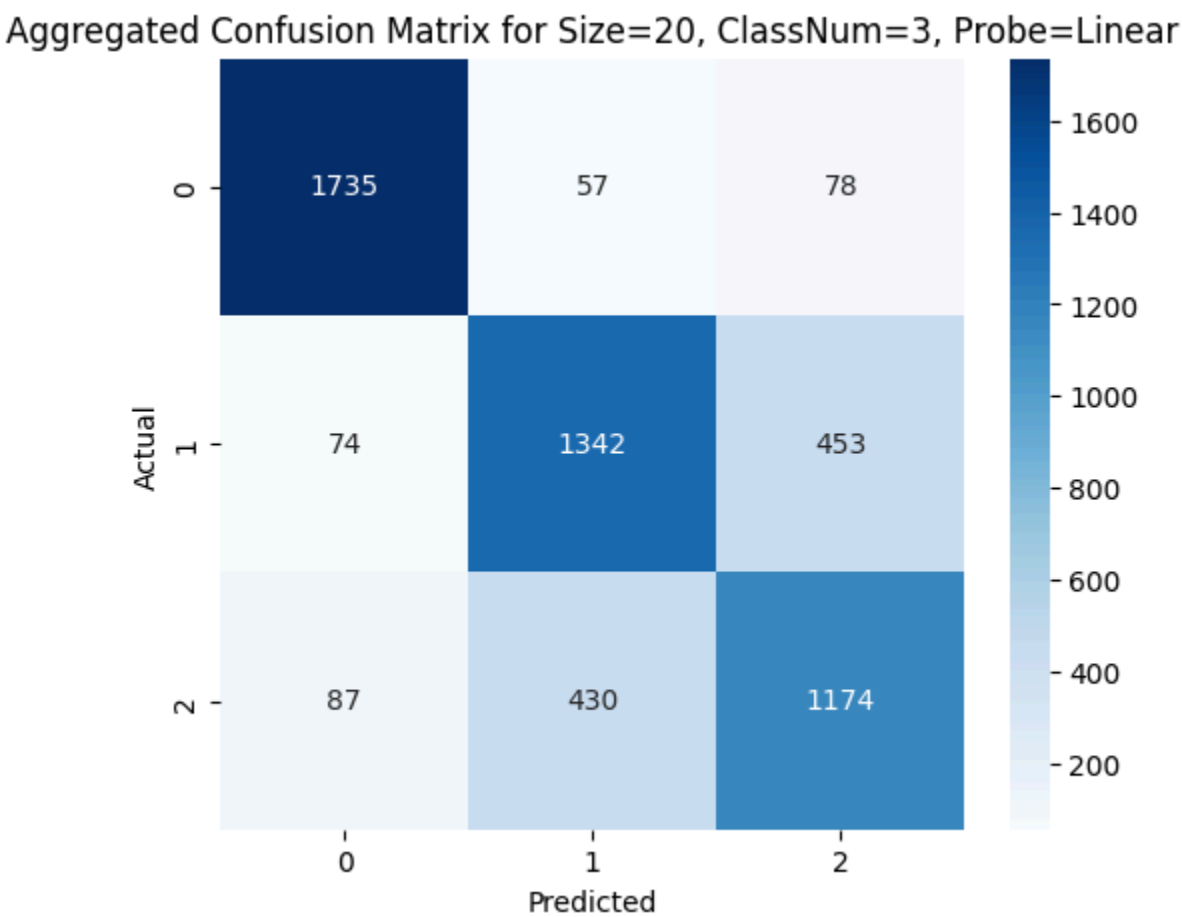


Size: 20

Train	Validation	Test
Accuracy: 0.8413 Loss: 0.3930	Accuracy: 0.7890 Loss: 0.4938	Accuracy: 0.7803 Loss: 0.5105

Precision	Recall	(Macro) F1 Score
0.7787	0.7777	0.7763

Confusion Matrix:



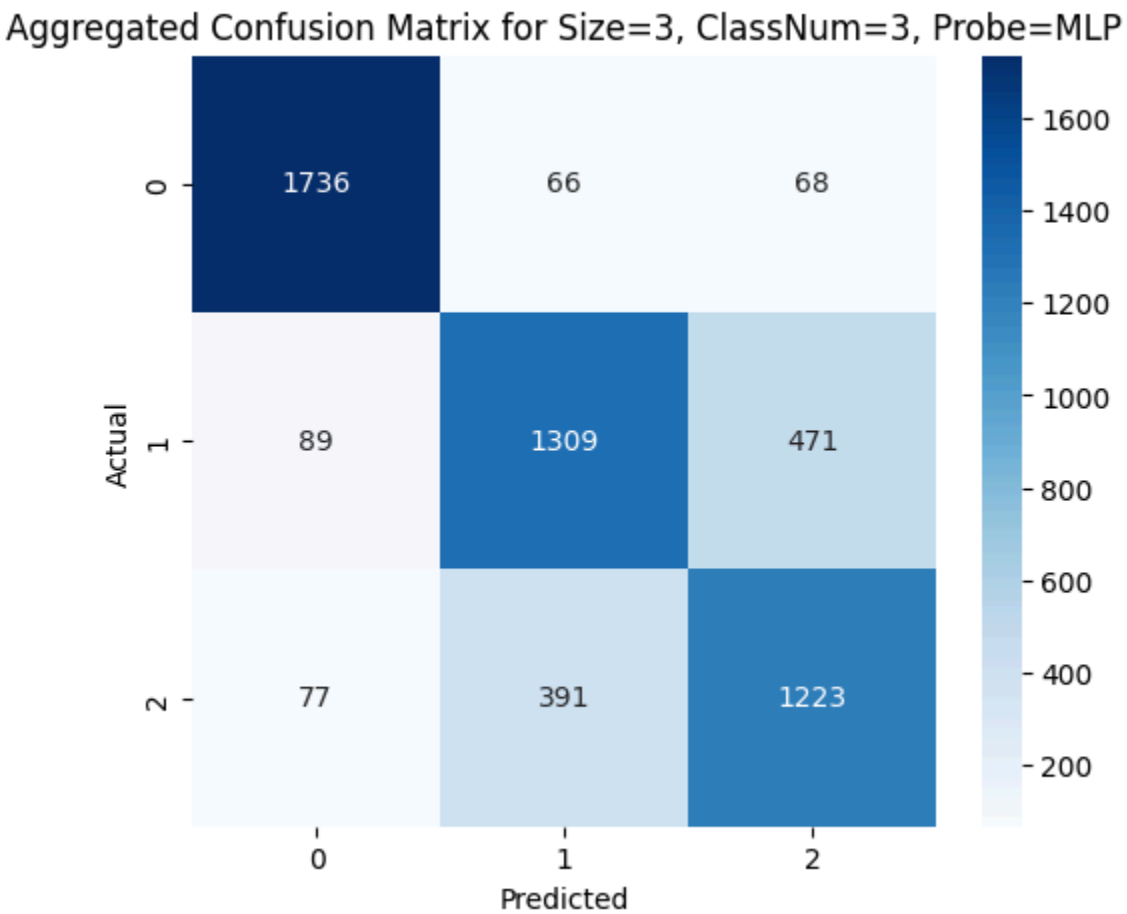
MLP Probe

Size: 3

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0032	Accuracy: 0.7835 Loss: 0.8543	Accuracy: 0.7817 Loss: 0.8868

Precision	Recall	(Macro) F1 Score
0.7803	0.7800	0.7781

Confusion Matrix:

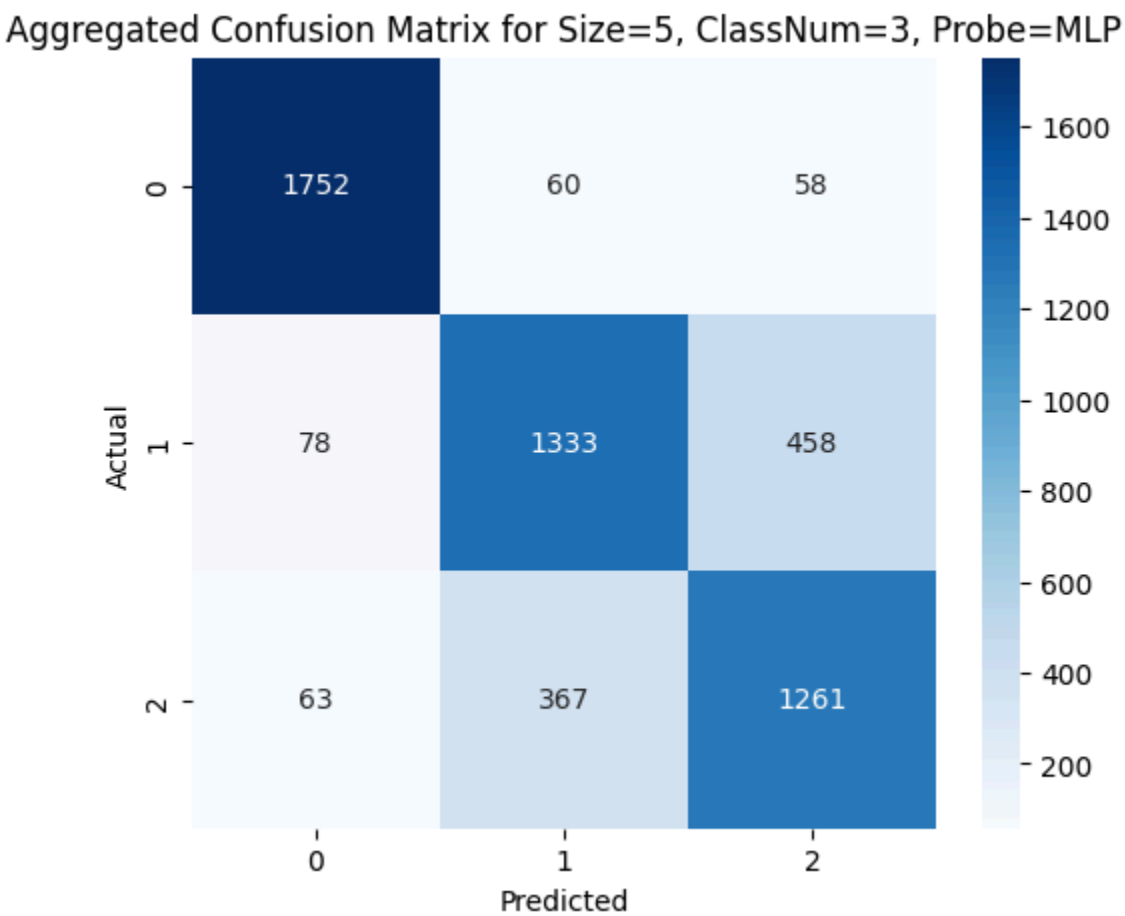


Size: 5

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0029	Accuracy: 0.7875 Loss: 0.8915	Accuracy: 0.7970 Loss: 0.9119

Precision	Recall	(Macro) F1 Score
0.7963	0.7955	0.7936

Confusion Matrix:

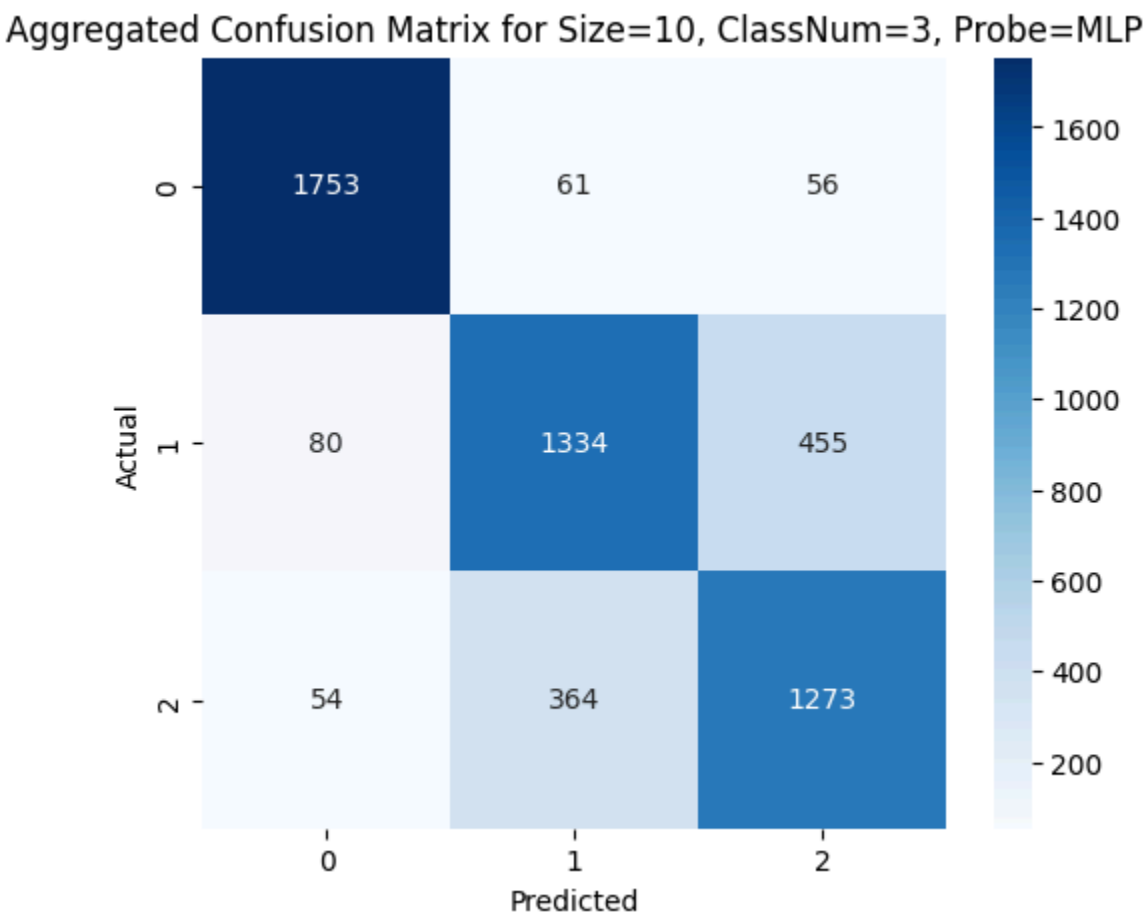


Size: 10

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0019	Accuracy: 0.8012 Loss: 0.9317	Accuracy: 0.8004 Loss: 0.9514

Precision	Recall	(Macro) F1 Score
0.7990	0.7989	0.7975

Confusion Matrix:

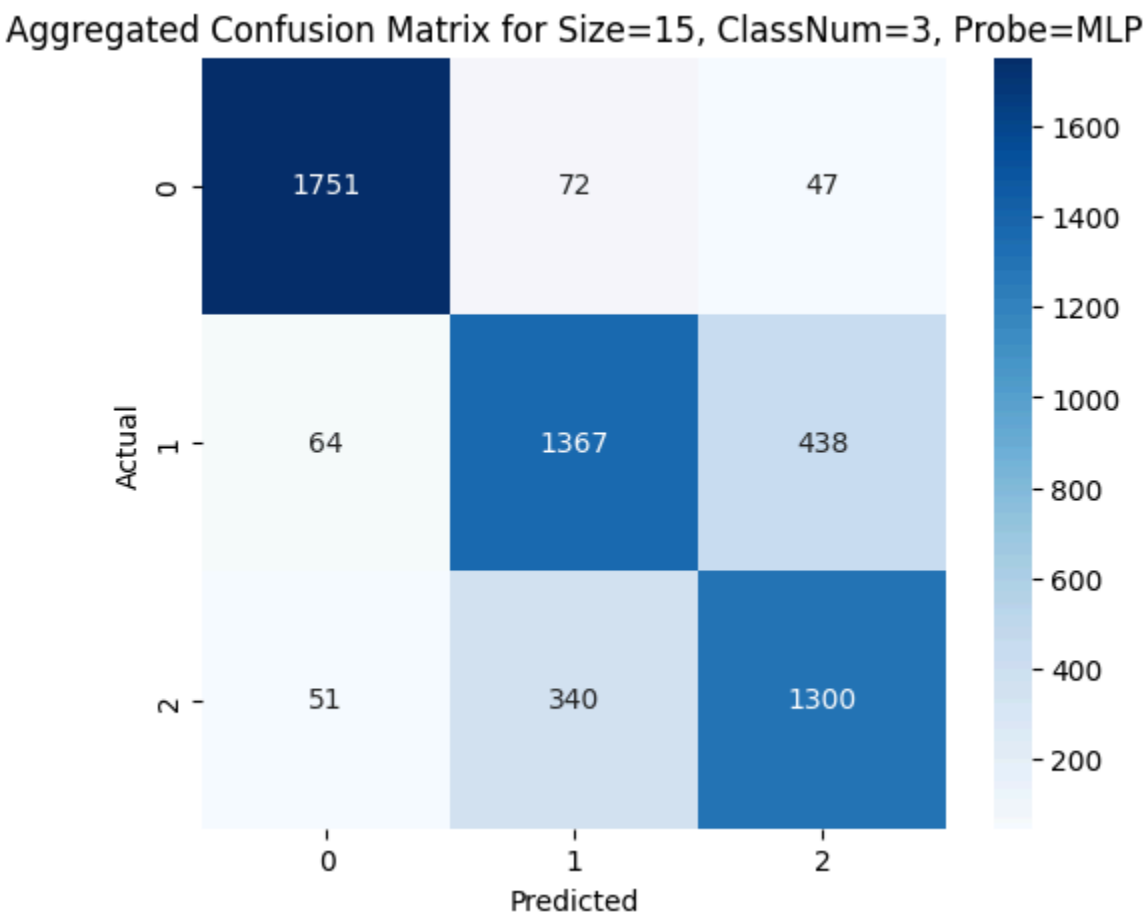


Size: 15

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0016	Accuracy: 0.8154 Loss: 0.9432	Accuracy: 0.8104 Loss: 0.9797

Precision	Recall	(Macro) F1 Score
0.8090	0.8090	0.8079

Confusion Matrix:

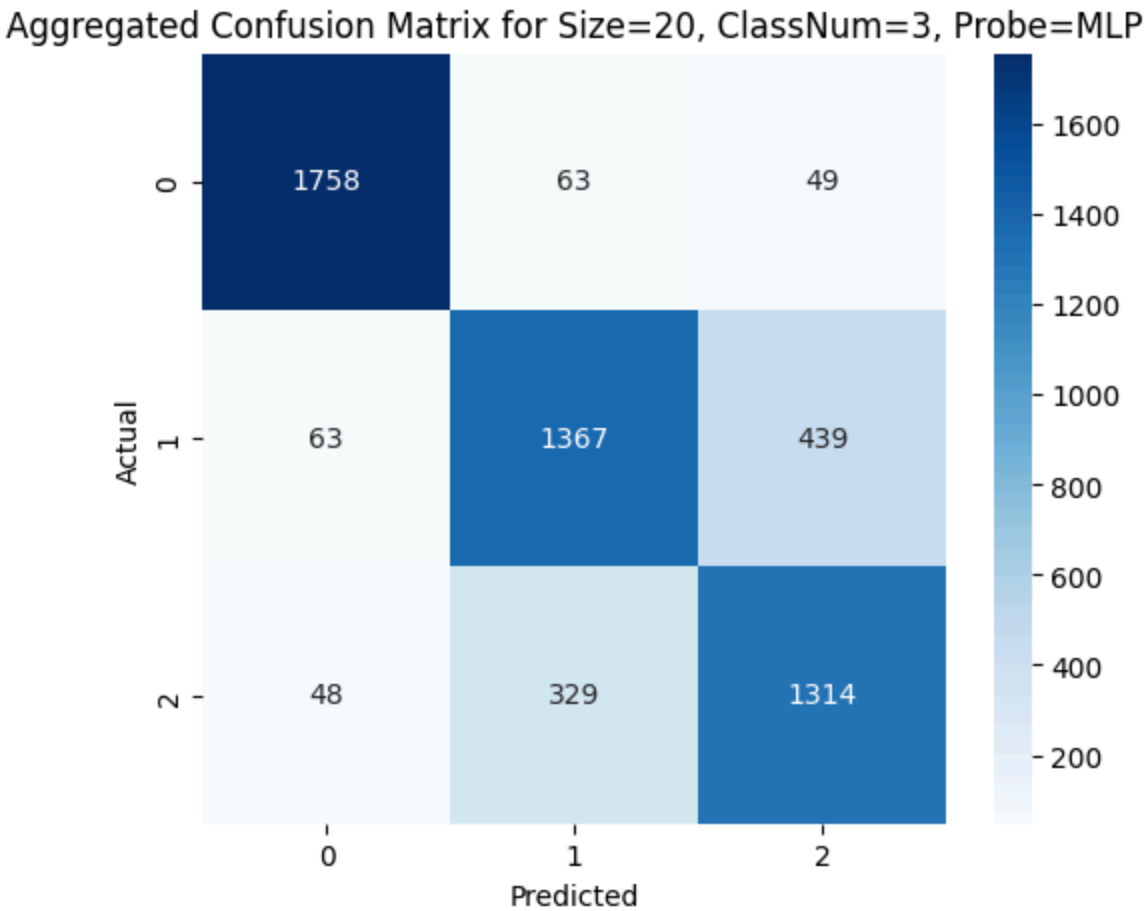


Size: 20

Train	Validation	Test
Accuracy: 1.0000 Loss: 0.0014	Accuracy: 0.8179 Loss: 0.9435	Accuracy: 0.8152 Loss: 0.9723

Precision	Recall	(Macro) F1 Score
0.8147	0.8141	0.8126

Confusion Matrix:



Task 3: ClassNum 9

Location classification: Which part of the face is being touched?

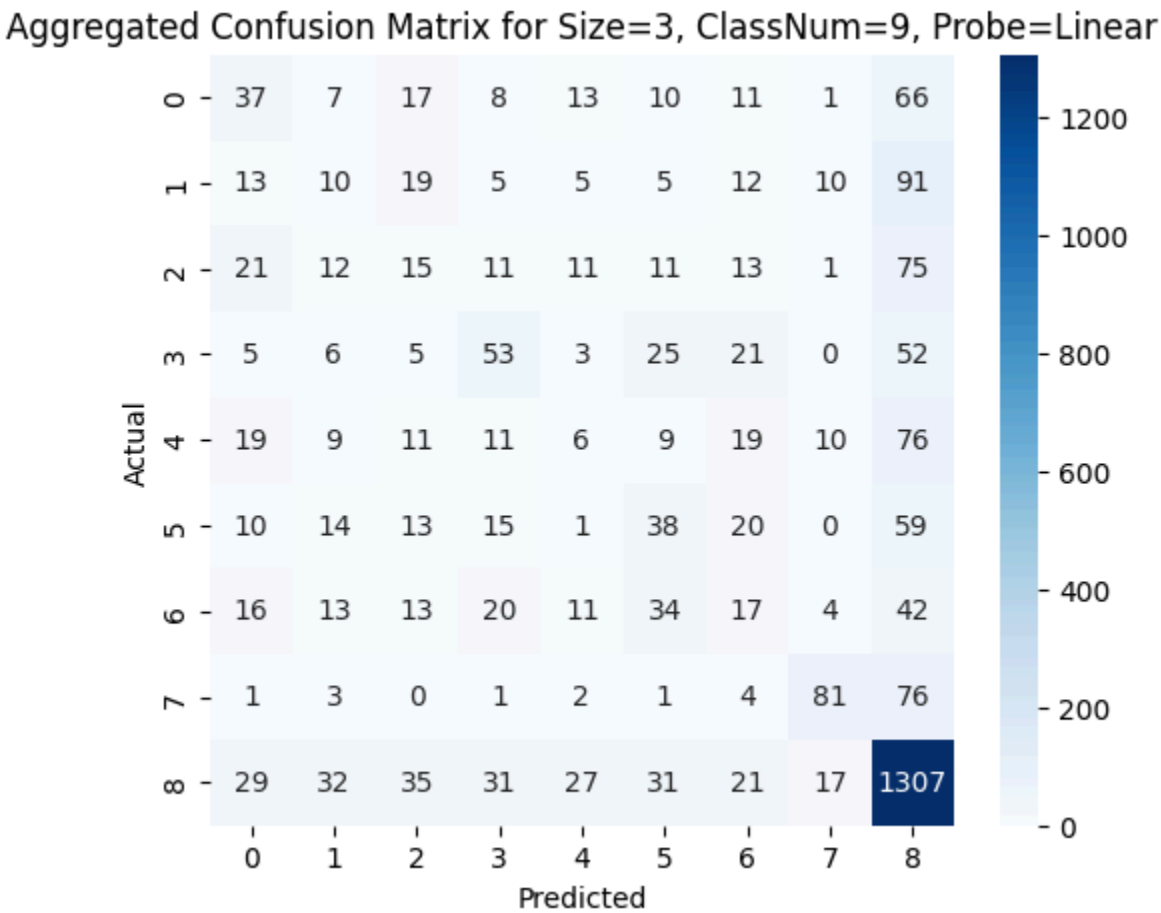
Linear Probe

Size: 3

Train	Validation	Test
Accuracy: 0.7823 Loss: 0.6675	Accuracy: 0.5360 Loss: 1.7067	Accuracy: 0.5395 Loss: 1.6856

Precision	Recall	(Macro) F1 Score
0.2793	0.2571	0.2592

Confusion Matrix:

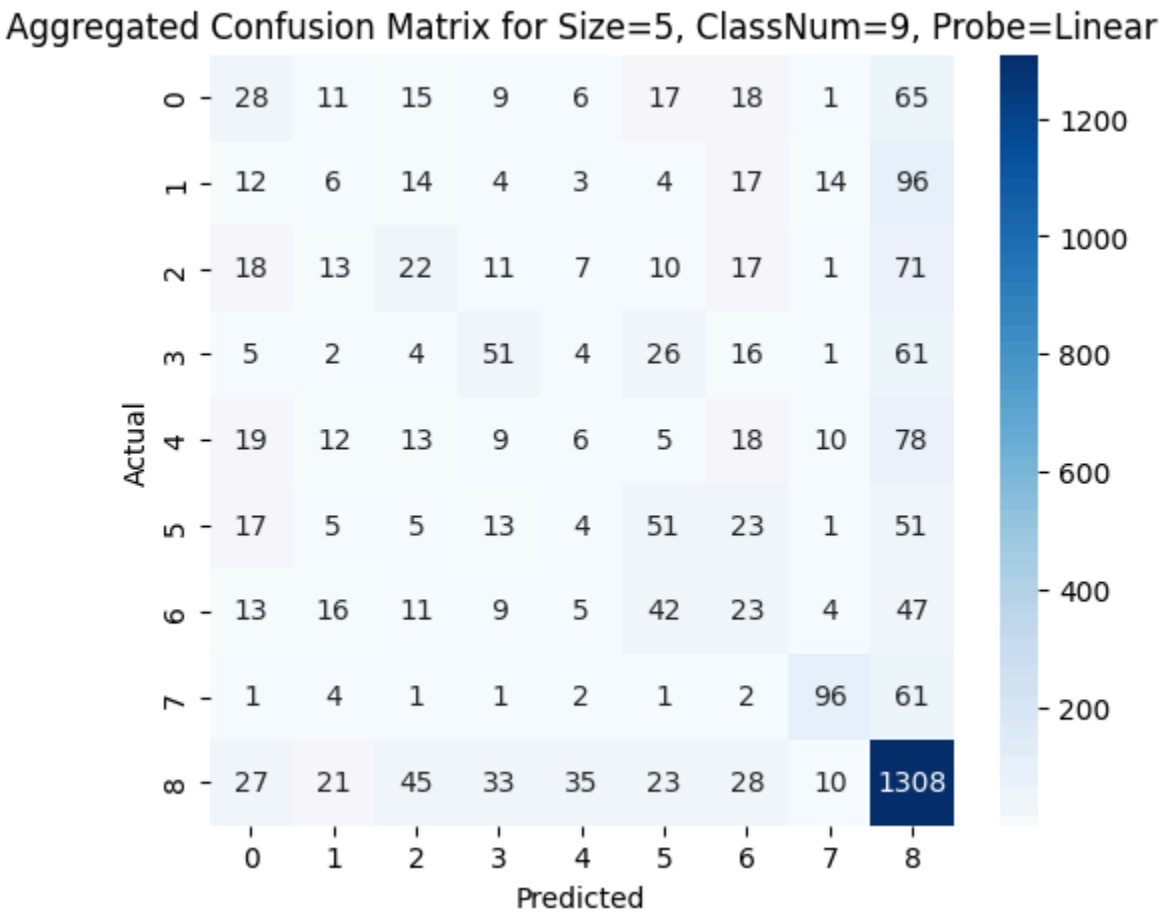


Size: 5

Train	Validation	Test
Accuracy: 0.7589 Loss: 0.7167	Accuracy: 0.5468 Loss: 1.6346	Accuracy: 0.5503 Loss: 1.6081

Precision	Recall	(Macro) F1 Score
0.3000	0.2767	0.2804

Confusion Matrix:

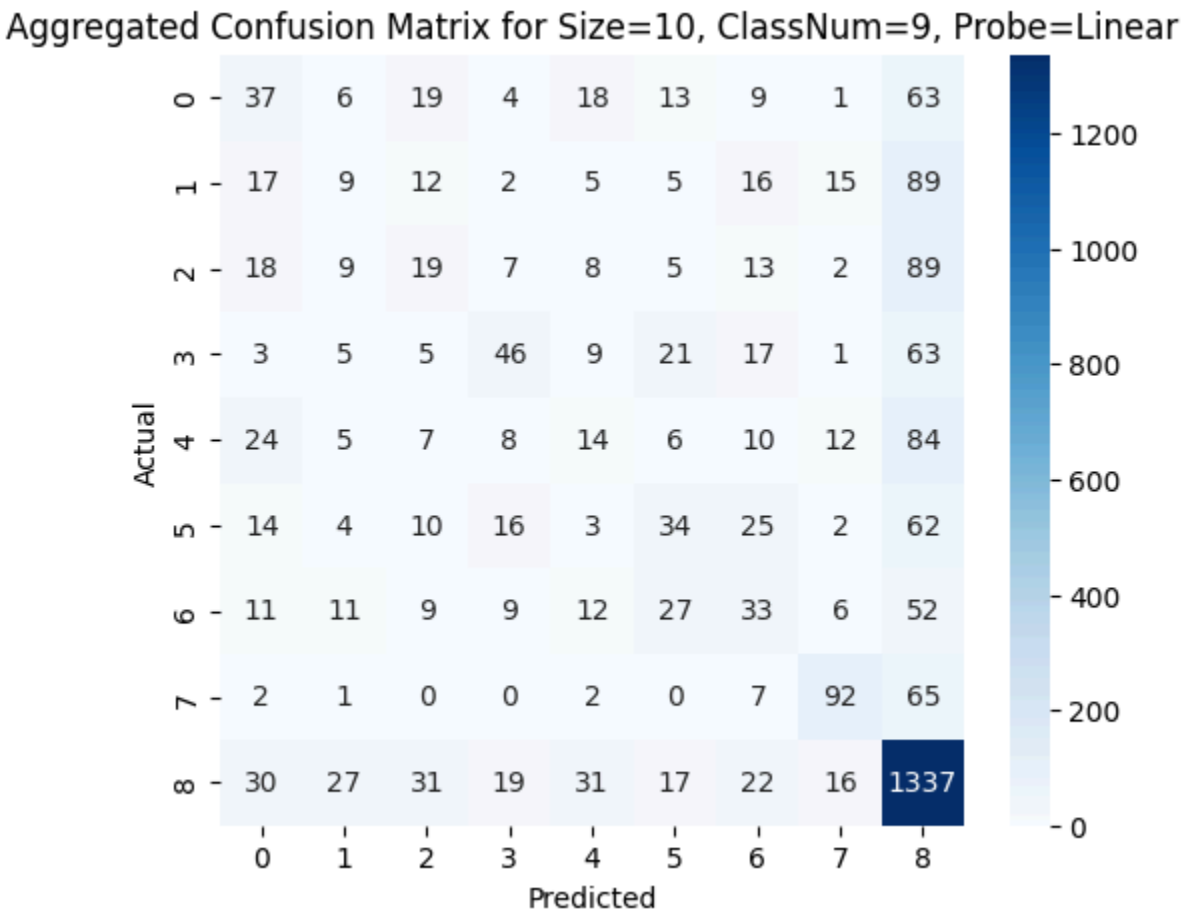


Size: 10

Train	Validation	Test
Accuracy: 0.7032 Loss: 0.8512	Accuracy: 0.5605 Loss: 1.5114	Accuracy: 0.5602 Loss: 1.4964

Precision	Recall	(Macro) F1 Score
0.3143	0.2766	0.2819

Confusion Matrix:

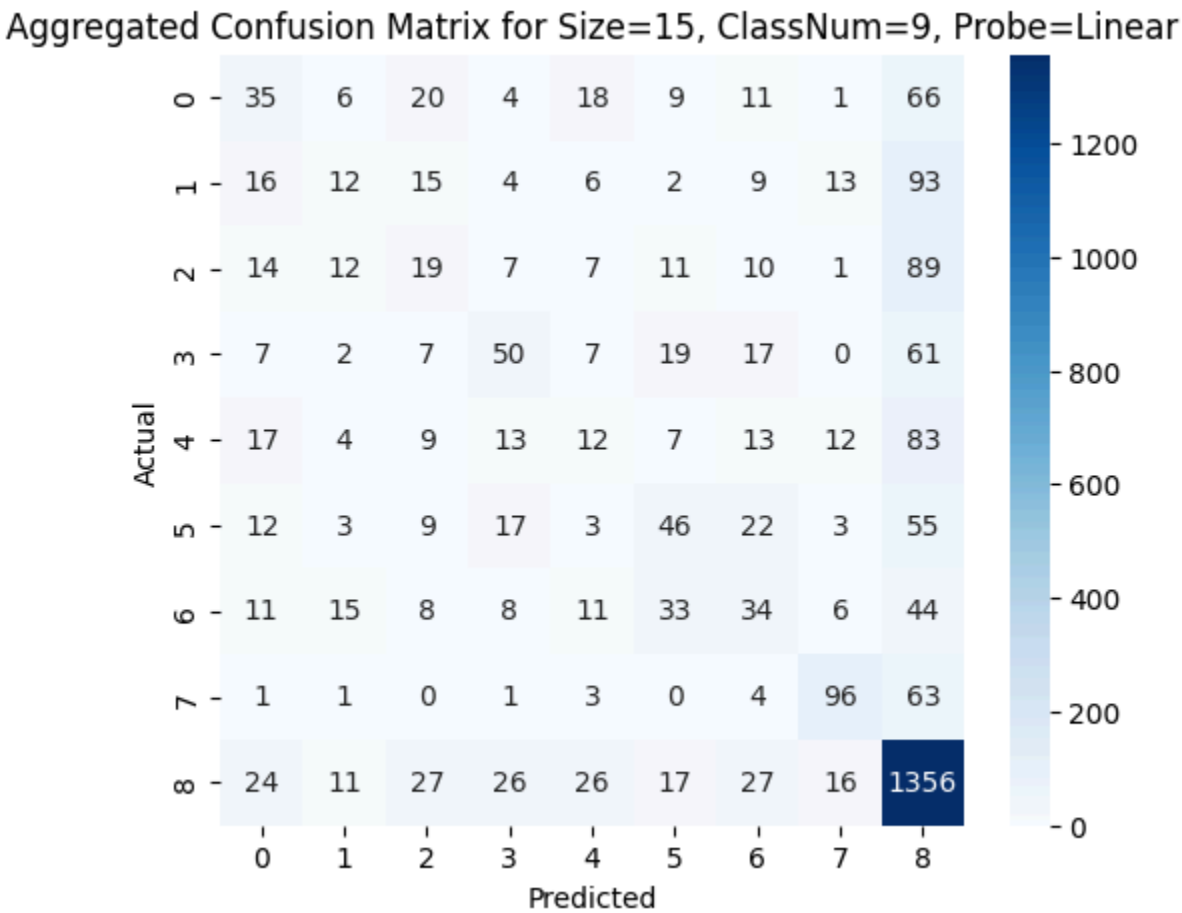


Size: 15

Train	Validation	Test
Accuracy: 0.6967 Loss: 0.8713	Accuracy: 0.5628 Loss: 1.4615	Accuracy: 0.5731 Loss: 1.4632

Precision	Recall	(Macro) F1 Score
0.3282	0.2914	0.2985

Confusion Matrix:

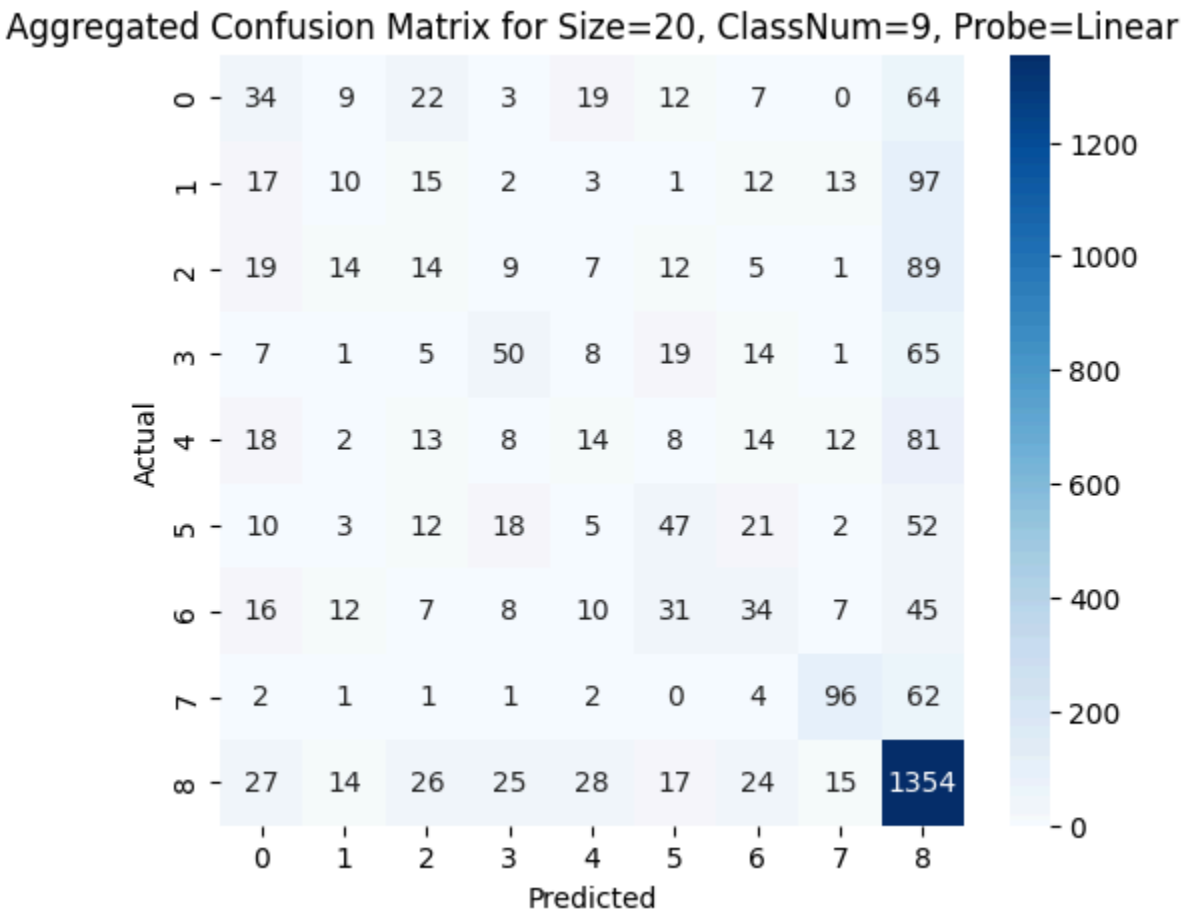


Size: 20

Train	Validation	Test
Accuracy: 0.6925 Loss: 0.8870	Accuracy: 0.5663 Loss: 1.4552	Accuracy: 0.5708 Loss: 1.4488

Precision	Recall	(Macro) F1 Score
0.3232	0.2882	0.2953

Confusion Matrix:



MLP Probe

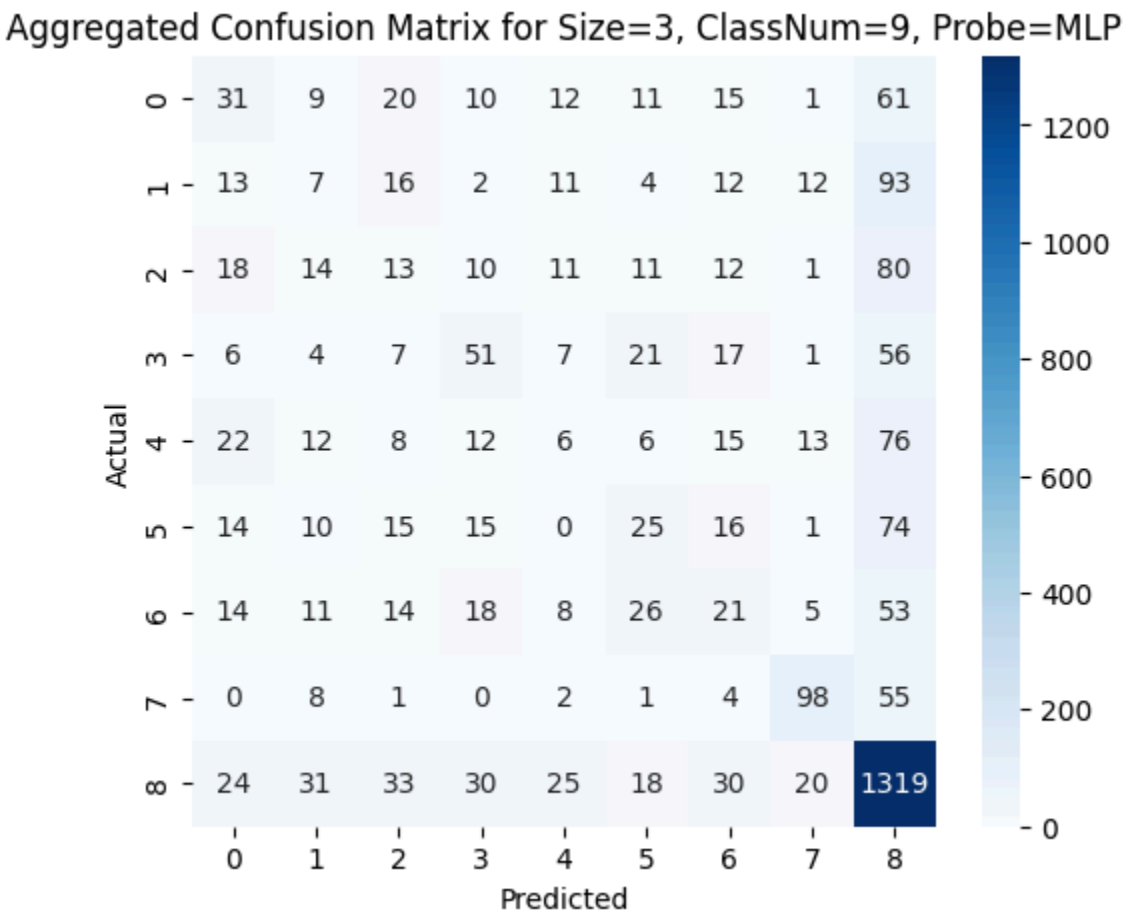
Size: 3

Train	Validation	Test
Accuracy: 1.0000	Accuracy: 0.5426	Accuracy: 0.5428

Loss: 0.0186	Loss: 2.4465	Loss: 2.3999

Precision	Recall	(Macro) F1 Score
0.2827	0.2564	0.2580

Confusion Matrix:



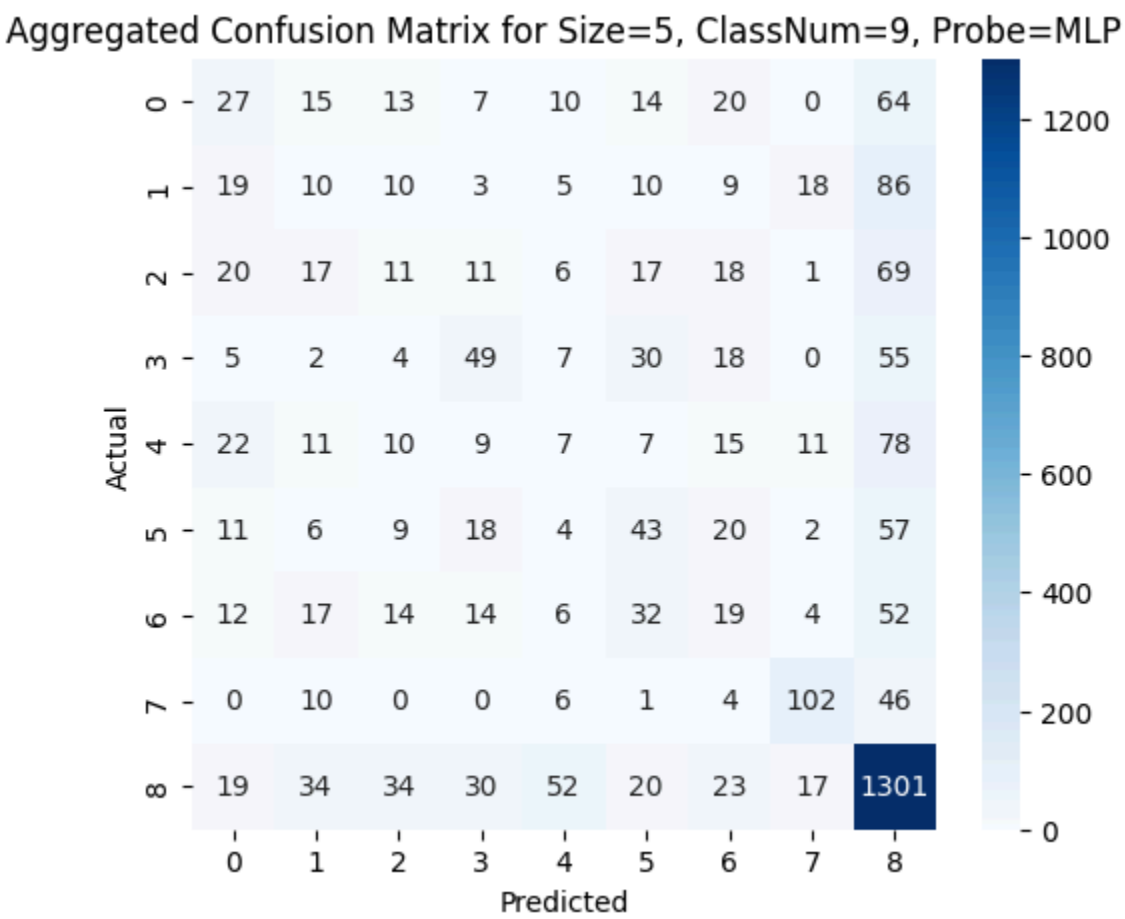
Size: 5

Train	Validation	Test

Accuracy: 1.0000 Loss: 0.0134	Accuracy: 0.5510 Loss: 2.5388	Accuracy: 0.5433 Loss: 2.4879

Precision	Recall	(Macro) F1 Score
0.2924	0.2659	0.2706

Confusion Matrix:



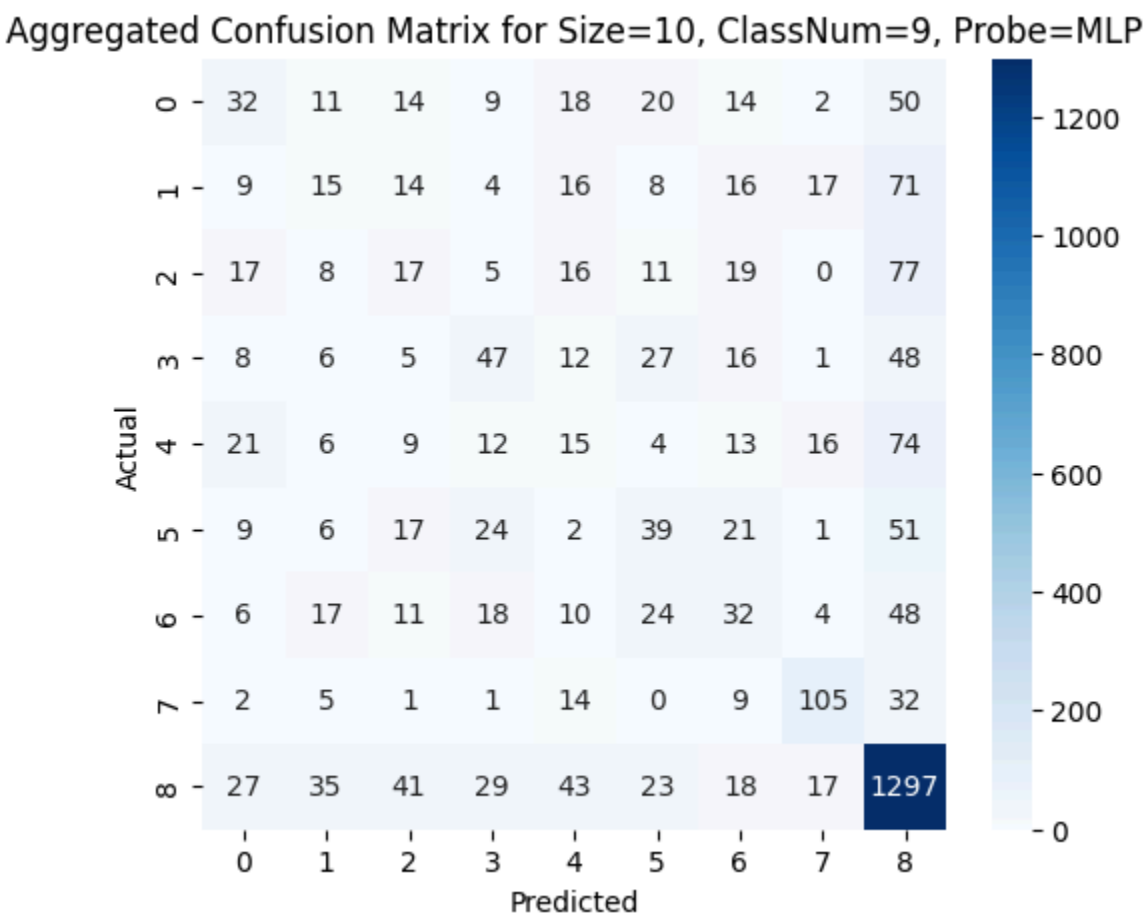
Size: 10

Train	Validation	Test

Accuracy: 1.0000 Loss: 0.0094	Accuracy: 0.5613 Loss: 2.7210	Accuracy: 0.5528 Loss: 2.7239

Precision	Recall	(Macro) F1 Score
0.3058	0.2860	0.2879

Confusion Matrix:



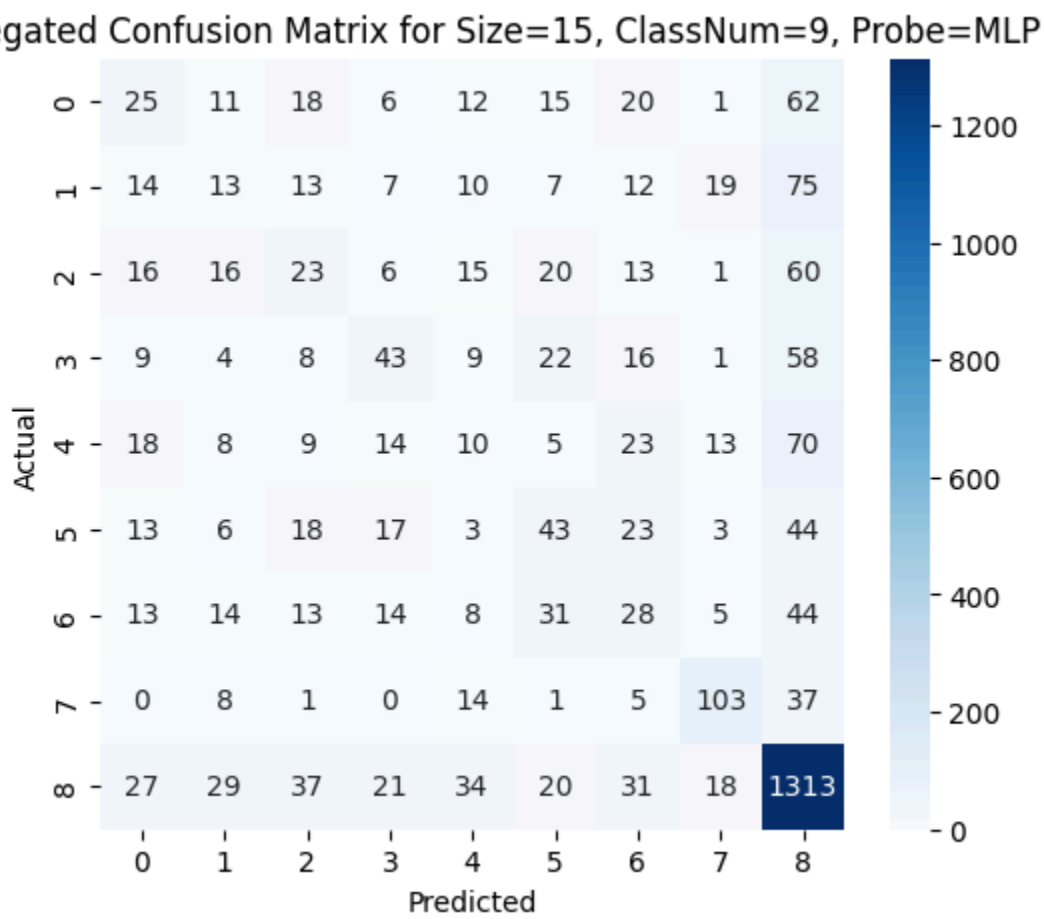
Size: 15

Train	Validation	Test

Accuracy: 1.0000 Loss: 0.0076	Accuracy: 0.5654 Loss: 2.9003	Accuracy: 0.5538 Loss: 2.8914

Precision	Recall	(Macro) F1 Score
0.3025	0.2797	0.2825

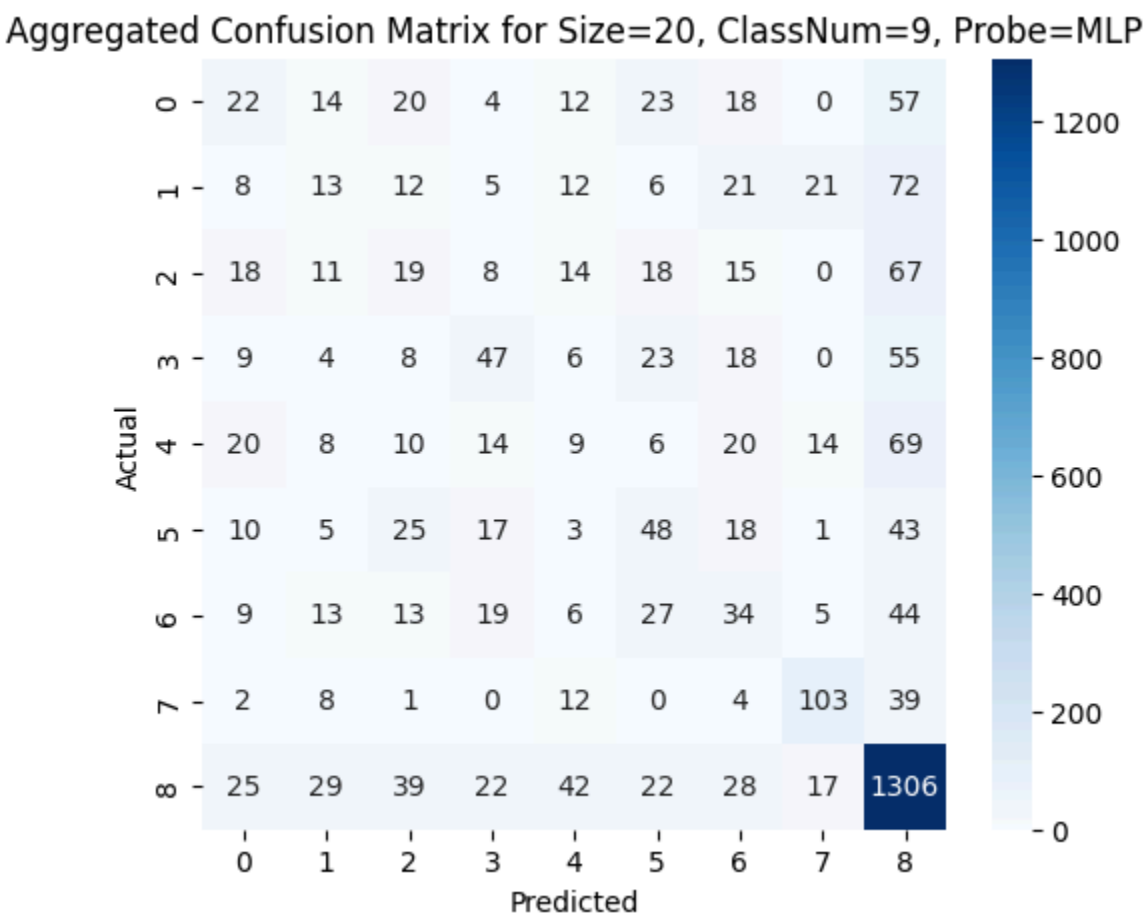
Confusion Matrix:



Size: 20

Train	Validation	Test

Accuracy: 1.0000 Loss: 0.0065	Accuracy: 0.5574 Loss: 3.0910	Accuracy: 0.5539 Loss: 3.0634

Precision	Recall	(Macro) F1 Score
0.2991	0.2825	0.2838

Confusion Matrix: